

The Cignetti and The Noise

Introduction

Hiring of a Division I football coach is among the most consequential decisions an Athletic Director and a University can make. Using data on 101 recent P4 college football hires, we built a statistical model for predicting a coach's success at their new school. For each hire, we collected data about their background and experiences, the previous success as a head coach or coordinator and their success since hiring. Over 50 variables on these factors were recorded, though we used 29 of these in building our predictive model.

Methods

We consider two novel measures of coaching success; each is based upon Bill Connelly's SP+ team ratings (measured in points above or below an average team) relative to the performance on the same metric of the school in the 15 years prior to their selection as head coach. The first is SP Rel, the raw difference in SP+ between each coach's tenure at a school and the average SP+ in the previous 15 years. The second, SP Rel_z is a normalized version to account for the difficulty of improving a team into the upper tiers of college football. Using a cross-validated regularized linear regression, we obtain a predictive model for coaching success.

Table 1: Top 10 and Bottom 10 Performances by *Sp Rel*.

name	school	SP Rel	SP Rel _z	Seasons
Curt Cignetti	Indiana	26.5	3.26	2
Josh Heupel	UCF	21.1	2.60	3
Sonny Dykes	SMU	17.8	2.18	5
Rhett Lashlee	SMU	17.0	2.09	4
Mike Gundy	Oklahoma State	13.8	1.69	21
Mike Elko	Duke	10.7	1.32	2
Dabo Swinney	Clemson	10.2	1.26	18
Matt Campbell	Iowa State	10.1	1.24	10
Bret Bielema	Illinois	9.3	1.14	5
Lane Kiffin	Ole Miss	9.2	1.13	6
⋮	⋮	⋮	⋮	⋮
Kalen DeBoer	Alabama	-11.5	-1.42	2
Ryan Walters	Purdue	-11.7	-1.44	2
Brent Pry	Virginia Tech	-13.0	-1.60	4
Deshaun Foster	UCLA	-13.2	-1.62	2
Jonathan Smith	Michigan State	-13.5	-1.67	2
Luke Fickell	Wisconsin	-14.8	-1.82	4
Geoff Collins	Georgia Tech	-16.3	-2.00	4
Chad Morris	Arkansas	-18.0	-2.21	3
Willie Taggart	Florida State	-19.1	-2.35	2
Troy Taylor	Stanford	-21.5	-2.65	2

Results

Among the important factors for predicting a successful hire are having been a previous college head coach, leaving a job as an Offensive Coordinator and quality of the hiring school's team in the previous 15 years to the hire. Specifically, the model indicates that having prior college head coaching experience or taking a job at a school with a higher quality in the last 15 years produces a negative effect on coaching success. This latter factor is a regression to the mean effect. However, the model also finds that leaving an offensive coordinator job has a positive impact on coaching success. No other factors were significant out of sample, see Figure 1 below.

While we do find these factors to be important for the prediction of a successful coaching hire, the trends are weak. With 66% accuracy, the model identifies coaching hires that will outperform team performance from the 15 years before the hire. Unsurprisingly, Curt Cignetti's performance (Table 1) as the coach at Indiana is an outlier. Although an outlier is not a trend, his performance is another example of unpredictability in college coaching performance.

Conclusion

No combination of these factors leads to high predictability of identifying a successful coaching hire. The data is noisy and presents mild patterns at best, indicating the many nuances of college football coaching. There are no magic coaching attributes that guarantee future success among the factors we investigated. Hiring a coach is hard; hiring a really successful coach is unlikely at best.

Figure 1: Plots for selected categorical predictors vs SP Rel

